

1 System Model and Problem Formulation

Batched LLM inference places two service objectives in direct tension: *system throughput*, the aggregate token rate delivered by the serving infrastructure (tokens/s), and *user interactivity*, the token rate experienced by each individual stream (tokens/s/user). Enlarging the verification batch improves throughput but dilutes per-user rates, while speculative decoding (SD) accelerates individual streams at the cost of verification compute that could otherwise serve additional users. In edge deployments this tension is sharpened by wireless round trips, heterogeneous draft–target agreement across users, and a hard roofline on the shared verification server. Existing SD serving systems—in the datacenter and at the edge—select a *single operating point* on this trade-off, e.g., maximum throughput, a fixed latency SLO, or a fairness utility. We instead treat the entire throughput–interactivity Pareto frontier as the design object and *maximize its area* (AUC). We formulate AUC maximization as a bilevel stochastic optimization coupling configuration (draft placement and sizing) with runtime control (speculation lengths, batching, and wireless resource allocation), show that the objective admits an operational robustness interpretation and a tractable dual structure, and lay out the structural, asymptotic, and online-learning results that the formulation supports.

1.1 Network and Speculative Decoding Model

We consider a set $\mathcal{N} = \{1, \dots, N\}$ of edge devices, each sustaining a persistent decode stream, and a shared edge server hosting the target model. Device i runs a draft model of size s_i selected from a library $\mathcal{D}$, with per-token drafting latency $\tau_i^d(s_i)$ and memory footprint $m(s_i)$. Time is slotted for control purposes; stream dynamics evolve in rounds. In round k , stream i drafts $\gamma_i(k) \in \{0, 1, \dots, \gamma_{\max}\}$ tokens, transmits $\kappa \gamma_i(k)$ bits uplink (token identifiers plus quantized draft probabilities, which suffice for lossless rejection-sampling verification) at rate

$$r_i(t) = w_i(t) \log_2(1 + \text{SNR}_i(t)), \quad \sum_{i \in \mathcal{N}} w_i(t) \leq W, \quad (1)$$

waits $W_i^q(k)$ in the server queue, is verified within a batch, and receives the accepted prefix plus one bonus token; δ_i collects the fixed per-round latency (downlink, propagation, protocol overhead). The server verifies a batch $\mathcal{B}$ carrying $\Gamma(\mathcal{B}) = \sum_{i \in \mathcal{B}} (\gamma_i + 1)$ tokens in one forward pass. Table 1 summarizes the notation.

We impose the following assumptions before the analysis.

Assumption 1 (Acceptance process). For each stream i , token-level acceptance events are i.i.d. Bernoulli(α_i) with $\alpha_i \in [\alpha_{\min}, \alpha_{\max}] \subset (0, 1)$, independent across streams; α_i is *unknown* to the controller. Consequently, a round with speculation length γ delivers A tokens (accepted prefix plus bonus) with

$$\mathbb{E}[A] = \phi(\gamma, \alpha) := \frac{1 - \alpha^{\gamma+1}}{1 - \alpha}, \quad (2)$$

and $A - 1$ is a geometric random variable truncated at γ .

Assumption 2 (Roofline verification). Verifying batch $\mathcal{B}$ takes

$$T^v(\mathcal{B}) = \theta_0 + \max \left\{ \underbrace{\frac{b_W + b_{KV} \sum_{i \in \mathcal{B}} L_i}{\beta_{\text{mem}}}}_{\text{memory-bound } \theta_m(\mathcal{B})}, \underbrace{\theta_f \Gamma(\mathcal{B})}_{\text{compute-bound}} \right\}, \quad (3)$$

with roofline knee $\Gamma^\dagger(\mathcal{B}) := \theta_m(\mathcal{B})/\theta_f$ at which the two regimes cross.

Table 1: Notation.

Symbol	Meaning
$\mathcal{N}, N$	set / number of edge streams
$\mathcal{D}, s_i, m(s_i)$	draft library, draft choice of stream i , its memory footprint
$\gamma_i(t), \gamma_{\max}$	speculation length, its cap
α_i	token-level acceptance rate of stream i (unknown)
$\phi(\gamma, \alpha)$	expected tokens delivered per round, (2)
$A_i(k)$	tokens delivered to stream i in round k (accepted + bonus)
τ_i^d	per-token drafting latency
κ	uplink payload per drafted token (bits)
$w_i(t), W, r_i(t)$	bandwidth allocation, budget, uplink rate (1)
δ_i	fixed per-round latency (downlink + overhead)
$\mathcal{B}(t), \Gamma(\mathcal{B})$	verification batch, its total token count
$\theta_0, \theta_f, \theta_m(\cdot)$	per-pass overhead, per-token compute time, memory floor
$b_W, b_{KV}, \beta_{\text{mem}}, L_i$	model bytes, KV bytes/token, memory bandwidth, context length
$\Gamma^\dagger$	roofline knee, (3)
$W_i^q(k)$	queueing delay of stream i in round k
x_i, Y	interactivity of stream i , system goodput
$\mathcal{C}(\xi), Y^*(x; \xi), \bar{x}(\xi)$	achievable region, frontier, max-min interactivity
$\text{AUC}(\xi), \widehat{\text{AUC}}(\xi)$	area under the frontier, its normalization
$\xi \in \Xi; \pi \in \Pi$	configuration; runtime policy
$Z_i(t), Q(t), Q_i(t)$	interactivity, server-compute, device-time virtual queues
$V, \lambda(t), \mu_i(t)$	Lyapunov weight, verification-token price, device-time price

Assumption 3 (Wireless channels). $\{\text{SNR}_i(t)\}$ are i.i.d. across slots (block fading), independent across streams, and induce bounded rates $0 < r_{\min} \leq r_i(t) \leq r_{\max}$ for any feasible allocation.

Assumption 4 (Closed-loop traffic). Each stream is persistent and closed-loop: round $k + 1$ is issued immediately upon receipt of round k 's response, so per-stream round epochs form renewal cycles under any stationary policy.

Assumption 5 (Drafting cost). Drafting γ tokens on device i takes $\gamma \tau_i^d$ seconds and is not overlapped with that stream's verification.

1.2 Interactivity, Goodput, and the Achievable Region

Let $\xi \in \Xi$ denote the *configuration*: the draft assignment $\{s_i\}_{i \in \mathcal{N}}$, the drafting placement (on-device vs. server-side), and the target replication/parallelism plan. Let $\pi \in \Pi$ denote the *runtime policy*: the speculation lengths $\gamma(t) = (\gamma_1(t), \dots, \gamma_N(t))$, the batch composition $\mathcal{B}(t)$, the bandwidth allocation $\mathbf{w}(t)$, and admission. Denote by $D_i(t)$ the cumulative tokens delivered to stream i by time t ; only accepted and bonus tokens count, so rejected drafts are wasted work that the control must manage.

Definition 1 (Interactivity and goodput). Under a stationary policy π , by the renewal-reward theorem (Assumption 4) the *interactivity* of stream i and the *system goodput* exist almost surely and satisfy

$$x_i(\pi, \xi) = \lim_{t \rightarrow \infty} \frac{D_i(t)}{t} = \frac{\phi(\gamma_i, \alpha_i)}{\mathbb{E}[\gamma_i \tau_i^d + \frac{\kappa \gamma_i}{r_i} + W_i^q + T^v + \delta_i]}, \quad Y(\pi, \xi) = \sum_{i \in \mathcal{N}} x_i(\pi, \xi). \quad (4)$$

Definition 2 (Achievable region and frontier). The achievable region, its frontier, and the max-min interactivity are

$$\mathcal{C}(\xi) := \overline{\text{conv}} \{ \mathbf{x}(\pi, \xi) : \pi \in \Pi \text{ stationary} \}, \quad Y^*(x; \xi) := \max \left\{ \sum_i x_i : \mathbf{x} \in \mathcal{C}(\xi), \min_i x_i \geq x \right\}, \quad (5)$$

for $x \in [0, \bar{x}(\xi)]$ with $\bar{x}(\xi) := \max \{ x : \exists \mathbf{x} \in \mathcal{C}(\xi), \min_i x_i \geq x \}$. Time-sharing between stationary policies attains the convex hull, so every point of $\mathcal{C}(\xi)$ is operational.

1.3 The AUC Objective and Problem Statement

Definition 3 (Area under the frontier). The AUC of configuration ξ and its normalization are

$$\text{AUC}(\xi) := \int_0^{\bar{x}(\xi)} Y^*(x; \xi) dx, \quad \widehat{\text{AUC}}(\xi) := \frac{\text{AUC}(\xi)}{\bar{x}(\xi) Y^*(0; \xi)} \in (0, 1], \quad (6)$$

where $Y^*(0; \xi)$ is the unconstrained maximum goodput. $\widehat{\text{AUC}}$ is unit-free and enables cross-system comparison.

We propose the following bilevel *frontier-area maximization* problem:

$$\begin{aligned} (\mathbf{P}) \quad & \max_{\xi \in \Xi} \text{AUC}(\xi) = \int_0^{\bar{x}(\xi)} Y^*(x; \xi) dx \\ & \text{s.t. } m(s_i) \leq M_i^{\text{dev}}, \quad \forall i \in \mathcal{N}, \\ & \quad b_W + b_{\text{KV}} K^{\text{max}}(\xi) + \sum_{s \in \text{server drafts}(\xi)} m(s) \leq M^{\text{srv}}, \end{aligned} \quad (7)$$

where $K^{\text{max}}(\xi)$ is the KV-cache budget implied by the admissible concurrency, and the inner value $Y^*(x; \xi)$ is the stochastic network optimization

$$\begin{aligned} (\mathbf{P}_x) \quad & Y^*(x; \xi) = \sup_{\pi \in \Pi} \liminf_{t \rightarrow \infty} \frac{1}{t} \sum_{i \in \mathcal{N}} \mathbb{E}[D_i(t)] \\ & \text{s.t. } \liminf_{t \rightarrow \infty} \frac{D_i(t)}{t} \geq x, \quad \forall i \in \mathcal{N}, \\ & \quad \text{all queues stable; } \sum_i w_i(t) \leq W; \quad \Gamma(\mathcal{B}(t)) \leq \Gamma_{\text{max}}, \quad \forall t. \end{aligned} \quad (8)$$

Decision variables and timescales. Problem (P) separates naturally into two timescales. The configuration ξ (draft sizing and placement, replication) is set at design time and *shapes* the frontier; the runtime policy π (speculation lengths, batching, bandwidth, admission) *operates* on it, and the admitted concurrency is the runtime knob that sweeps the operating point along the curve. All results below hold for fixed ξ ; the outer problem is a combinatorial selection over a profiled or learned surrogate of $\text{AUC}(\cdot)$.

1.4 AUC Is the Right Objective

Two results justify the objective in (7) operationally and computationally: the first shows that AUC maximization is exactly robust design against unknown interactivity requirements; the second shows the frontier—and hence the integral—is computable by a one-dimensional dual sweep.

Proposition 1 (Operational meaning of AUC). *Fix ξ and let w be any probability density on $[0, \bar{x}(\xi)]$. Then $Y^*(\cdot; \xi)$ is nonincreasing and bounded, hence measurable, and*

$$\int_0^{\bar{x}(\xi)} w(x) Y^*(x; \xi) dx = \mathbb{E}_{X \sim w} [Y^*(X; \xi)], \quad \text{AUC}(\xi) = \bar{x}(\xi) \cdot \mathbb{E}_{X \sim \text{Unif}[0, \bar{x}(\xi)]} [Y^*(X; \xi)]. \quad (9)$$

Hence the AUC-optimal configuration is the Bayes-optimal design when the per-user interactivity requirement is random or unknown with law w ; single-point goodput or SLO designs are recovered as the degenerate case $w = \delta_{x_0}$.

Proof. Monotonicity of $Y^*(\cdot; \xi)$ follows since the feasible set of (5) shrinks in x ; boundedness follows from bounded per-round rewards and durations (Assumptions 3–5, $\gamma_i \leq \gamma_{\max}$). A bounded monotone function is Borel measurable and Riemann integrable, and (9) is then the definition of expectation; the second identity is the uniform special case. $\square$

Proposition 2 (Concavity and dual sweep). *For every ξ : (i) $\mathcal{C}(\xi)$ is convex and compact; (ii) $Y^*(\cdot; \xi)$ is concave, nonincreasing, and continuous on $[0, \bar{x}(\xi)]$; (iii) for each $x \in [0, \bar{x}(\xi)]$ strong duality holds for (P_x) , and every frontier point is attained by a stationary randomized policy solving a weighted problem $\max_{\mathbf{x} \in \mathcal{C}(\xi)} \sum_i (1 + \eta_i) x_i$ for some multiplier vector $\boldsymbol{\eta} \in \mathbb{R}_+^N$; (iv) consequently the entire frontier, and the integral $\text{AUC}(\xi)$, are traced by a one-parameter dual path, i.e., a single max-weight algorithm evaluated along a sweep.*

Proof sketch. (i) Randomized time-sharing between stationary policies convexifies the rate set; compactness follows from the bounds in Proposition 1’s proof. (ii) For $x = \theta x' + (1 - \theta)x''$, the convex combination of optimizers for x' and x'' lies in $\mathcal{C}(\xi)$ and satisfies the min-rate constraint at x , giving $Y^*(x) \geq \theta Y^*(x') + (1 - \theta)Y^*(x'')$; monotonicity is immediate and concavity implies continuity on the interior. (iii) The constraint $\min_i x_i \geq x$ is the intersection of N linear constraints; Slater’s condition holds for $x < \bar{x}(\xi)$, so KKT multipliers $\boldsymbol{\eta}(x)$ exist and support the frontier point. (iv) Parametrize by the support-plane slope; concavity makes the sweep monotone and the integral a line integral along the dual path. $\square$

Remark 1 (Relation to GoodSpeed). The proportional-fair goodput allocation of our prior work GoodSpeed [?] solves $\max_{\mathbf{x} \in \mathcal{C}(\xi)} \sum_i \log x_i$ and therefore selects a *single* supported point of $\partial \mathcal{C}(\xi)$. Problem (P) strictly generalizes it: the log-utility point is one tangency of the dual sweep in Proposition 2(iv), while (P) optimizes the curve on which that point lives.

1.5 Structural Characterization of the Frontier

We first characterize the frontier in the symmetric saturated instance, which isolates the mechanism generating the Pareto curvature: $\alpha_i = \alpha$, $\tau_i^d = \tau^d$, $r_i = r$, $\delta_i = \delta$, $b_{KV} = 0$ (so θ_m is constant), and gated batch service in which the server synchronously verifies the B admitted streams at a common speculation length γ (so $W_i^d = 0$). The per-round cycle time and the resulting operating point are

$$T(B, \gamma) = \gamma \left(\tau^d + \frac{\kappa}{r} \right) + \delta + \theta_0 + \max \{ \theta_m, \theta_f B(\gamma + 1) \}, \quad x(B, \gamma) = \frac{\phi(\gamma, \alpha)}{T(B, \gamma)}, \quad Y = B x(B, \gamma). \quad (10)$$

Theorem 1 (Two-regime frontier). *Under Assumptions 1–5 in the symmetric saturated instance, with $\Delta T(B, \gamma) := T(B, \gamma + 1) - T(B, \gamma)$:*

- (a) $\gamma \mapsto \phi(\gamma, \alpha)$ is strictly increasing and discretely concave with increments $\phi(\gamma + 1, \alpha) - \phi(\gamma, \alpha) = \alpha^{\gamma+1}$;

(b) for each B , $\gamma \mapsto x(B, \gamma)$ is unimodal, and

$$\gamma^*(B) = \max\left\{\gamma : \alpha^{\gamma+1} T(B, \gamma) \geq \phi(\gamma, \alpha) \Delta T(B, \gamma)\right\}, \quad (11)$$

i.e., speculate while the relative acceptance gain exceeds the relative cycle-time cost;

- (c) $\gamma^*(B)$ is nonincreasing in B , and along the frontier $\gamma^*(x)$ is nondecreasing in x : speculation deepens as the operating point moves toward the interactivity end;
- (d) (SD-off threshold) there exists a finite B_c (equivalently $x_c > 0$) with $\gamma^*(B) = 0$ for all $B \geq B_c$: in the server-bottlenecked compute-bound regime the aggregate goodput equals $\phi(\gamma, \alpha)/((\gamma + 1)\theta_f)$, and $\phi(\gamma, \alpha)/(\gamma + 1) < 1 = \phi(0, \alpha)$ for all $\gamma \geq 1$ and $\alpha < 1$, so speculation is strictly suboptimal;
- (e) (RTT amortization) $\gamma^*(B)$ is nondecreasing in the fixed per-round latency $\delta + \theta_0$ and nonincreasing in the per-token cost $\tau^d + \kappa/r$; high-RTT edge links optimally speculate deeper at any fixed load.

Consequently $Y^*(\cdot)$ is concave and piecewise smooth with at most two regimes—memory-bound (high x , verification nearly free, speculation amortizes the wireless round trip) and compute-bound (low x , every drafted token cannibalizes batch capacity)—joined at the roofline knee $\Gamma^\dagger$ of (3).

Proof sketch. (a) is direct from (2). (b): $x(B, \gamma + 1) \geq x(B, \gamma)$ iff $\alpha^{\gamma+1} T(B, \gamma) \geq \phi(\gamma, \alpha) \Delta T(B, \gamma)$; the left ratio $\alpha^{\gamma+1}/\phi(\gamma, \alpha)$ is strictly decreasing in γ while $\Delta T/T$ is nonincreasing-then-constant, yielding unimodality and the stopping rule (11), a discrete Dinkelbach condition. (c) and (e) follow from monotone comparative statics (Topkis) applied to the log-transformed ratio: B and $\tau^d + \kappa/r$ raise the marginal cost side of (11) while $\delta + \theta_0$ raises only the amortizable fixed cost; composing with the monotone map $x \mapsto B^*(x)$ gives the frontier statement. (d): in the saturated compute-bound regime the server is the bottleneck and delivers $\phi(\gamma, \alpha)$ useful tokens per $(\gamma + 1)\theta_f$ seconds of verification; the token-efficiency bound $\phi(\gamma, \alpha) \leq \gamma + 1$ with equality only at $\gamma = 0$ (for $\alpha < 1$) forces $\gamma^* = 0$. Concavity of the frontier is inherited from Proposition 2(ii). $\square$

Remark 2 (Edge-specific tension). Parts (d) and (e) formalize the two opposing forces that make the *distributed* problem qualitatively different from datacenter SD: wireless round-trip amortization pushes γ up while verification contention pushes it down, so the optimal policy is spatially heterogeneous even for identical acceptance rates, and the sign of the net effect flips across the frontier.

Theorem 2 (Fluid limit and finite- N gap). *Consider the sequence of systems indexed by N with server token rate scaled as $c_N = cN$ and i.i.d. stream parameters. Then the empirical measure of stream states converges, uniformly on compacts and in probability, to the unique fixed point of a fluid model; $Y_N^*(x)/N \rightarrow y^*(x)$ for a closed-form limiting frontier y^* ; and $\sup_x |Y_N^*(x)/N - y^*(x)| = O(1/\sqrt{N})$.*

Proof strategy. The closed system of N circulating rounds with think times (drafting plus uplink) and a roofline batch server is a density-dependent Markov chain; Kurtz’s theorem and exchangeability (propagation of chaos) give the fluid limit, and a functional CLT gives the $O(1/\sqrt{N})$ rate. This extends the fluid sample-path analysis of GoodSpeed [?] from a single log-utility operating point to the parametric family $\{(P_x)\}_{x \in [0, \bar{x}]}$. $\square$

1.6 Online Distributed Control with Censored Learning

We now solve (P_x) online, without knowledge of $\{\alpha_i\}$, by a drift-plus-penalty controller whose per-slot subproblems decouple across devices, server, and radio. Let Δ_s be the slot length and $a_i(t)$ the tokens delivered to stream i in slot t . Define the interactivity, server-compute, and device-time virtual queues

$$Z_i(t+1) = [Z_i(t) + x\Delta_s - a_i(t)]^+, \quad Q(t+1) = [Q(t) + \theta_f \Gamma(\mathcal{B}(t)) - \Delta_s]^+, \quad Q_i(t+1) = \left[Q_i(t) + \gamma_i(t) \left(\tau_i^d + \frac{\kappa}{r_i(t)} \right) - \Delta_s \right]^+, \quad (12)$$

and the prices $\lambda(t) := Q(t)$ and $\mu_i(t) := Q_i(t)$. Minimizing the one-slot drift-plus-penalty bound decouples into three subproblems.

(i) Device-side speculation index. The marginal benefit of the $(\gamma+1)$ -th draft token is $(V + Z_i(t)) \alpha_i^{\gamma+1}$ expected deliverable tokens, while its marginal cost is $\lambda(t)\theta_f + \mu_i(t)(\tau_i^d + \kappa/r_i(t))$; optimal stopping yields the closed form

$$\gamma_i(t) = \left[\left\lceil \frac{\ln c_i(t)}{\ln \hat{\alpha}_i(t)} \right\rceil - 1 \right]_0^{\gamma_{\max}}, \quad c_i(t) = \frac{\lambda(t)\theta_f + \mu_i(t)(\tau_i^d + \kappa/r_i(t))}{V + Z_i(t)}, \quad (13)$$

where $[\cdot]_0^{\gamma_{\max}}$ clamps to $\{0, \dots, \gamma_{\max}\}$ and $\hat{\alpha}_i(t)$ is the current (optimistic) acceptance estimate. The server broadcasts the single scalar $\lambda(t)$ —a posted *verification-token price*—and devices respond selfishly, so the architecture is one-scalar-feedback network utility maximization.

(ii) Server-side batching. Batch selection is the token-budget knapsack $\max_{\mathcal{S}} \sum_{i \in \mathcal{S}} \omega_i(t)$ s.t. $\Gamma(\mathcal{S}) \leq \Gamma_{\text{bud}}(t)$ with weights $\omega_i(t) = (V + Z_i(t)) \phi(\gamma_i, \hat{\alpha}_i)$ over pending verifications; the greedy rule ranked by $\omega_i/(\gamma_i + 1)$ is optimal for the LP relaxation and within one job of integer-optimal, with $\Gamma_{\text{bud}}(t)$ set to the knee $\Gamma^\dagger$ (memory-bound preserving) or driven by $Q(t)$.

(iii) Radio allocation. Given per-stream time prices, minimizing $\sum_i \mu_i(t) \kappa \gamma_i(t) / (w_i g_i(t))$ with $g_i(t) = \log_2(1 + \text{SNR}_i(t))$ subject to $\sum_i w_i \leq W$ admits the square-root waterfilling solution $w_i^*(t) = W \sqrt{\mu_i \kappa \gamma_i / g_i} / \sum_j \sqrt{\mu_j \kappa \gamma_j / g_j}$ by Cauchy–Schwarz.

Theorem 3 (Controller optimality). *Under Assumptions 1–5 with known $\{\alpha_i\}$, for any $V > 0$ the controller (12)–(13) achieves*

$$\liminf_{t \rightarrow \infty} \frac{1}{t} \sum_{\tau \leq t} \mathbb{E} \left[\sum_i a_i(\tau) \right] \geq Y^*(x; \xi) - \frac{C}{V}, \quad \mathbb{E}[Z_i(t) + Q(t) + Q_i(t)] = O(V) \quad \forall i, \quad (14)$$

so every stream’s time-average interactivity satisfies its constraint within $O(V/t)$, for a constant C determined by the bounded second moments of per-slot rewards and durations.

Proof sketch. Quadratic Lyapunov function on $(\mathbf{Z}, Q, \mathbf{Q})$; the per-slot minimizer of the drift-plus-penalty bound is exactly (i)–(iii); boundedness of $\gamma_{\max}$, $r_{\max}$, and (3) gives the constant C ; the $[O(1/V), O(V)]$ trade-off follows from the standard telescoping argument, and Proposition 2(iii) guarantees a stationary randomized policy achieving $Y^*(x; \xi)$ to compare against. $\square$

Theorem 4 (Censored learning of acceptance rates). *Let $\{\alpha_i\}$ be unknown in $[\alpha_{\min}, \alpha_{\max}]$. Each round with $\gamma_i \geq 1$ reveals a truncated-geometric observation: the accepted prefix contributes Bernoulli*

successes and, unless all γ_i drafts are accepted (right-censoring), one failure. Running (13) with upper-confidence estimates $\hat{\alpha}_i^{\text{ucb}}(t)$ built from Bernstein confidence intervals on these samples yields

$$\text{Reg}(T) := TY^*(x; \xi) - \sum_{t \leq T} \mathbb{E} \left[\sum_i a_i(t) \right] = \tilde{O}(\sqrt{NT}) + O(T/V), \quad (15)$$

while preserving the $O(V)$ backlog guarantees of Theorem 3.

Proof sketch. Each informative round contributes at least one Bernoulli sample, so $|\hat{\alpha}_i - \alpha_i| = \tilde{O}(1/\sqrt{n_i})$ by Bernstein; the index (13) and $\phi(\cdot, \alpha)$ are Lipschitz in α on the compact $[\alpha_{\min}, \alpha_{\max}]$; optimism bounds per-round regret by the confidence width, and summing $\sqrt{1/n_i}$ over streams and rounds gives $\tilde{O}(\sqrt{NT})$; the $O(T/V)$ term is the controller gap of Theorem 3. $\square$

Remark 3 (Self-exploring constraint pressure). Rounds with $\gamma_i = 0$ are uninformative, an earning-while-learning obstruction absent from unconstrained bandit formulations. The controller resolves it endogenously: whenever stream i 's interactivity constraint binds, $Z_i(t)$ grows, $c_i(t) \downarrow 0$ in (13), and $\gamma_i(t) \geq 1$ is forced—constraint pressure guarantees exploration, and explicit forced exploration is needed only for streams with slack constraints.

1.7 Extensions

Remark 4 (Multi-server routing via posted prices). With M verification servers each posting a price $\lambda_m(t)$, devices jointly select (m, γ) by the natural extension of (13). By Proposition 2(i) the aggregate problem remains convex, so the posted-price equilibrium coincides with the dual optimum and routing incurs no efficiency loss; the transient is a congestion-game dynamic amenable to standard analysis.

Remark 5 (Hierarchical draft cascades). A device tiny-draft $\rightarrow$ mid-tier verify-and-redraft $\rightarrow$ target cascade is a tandem of speculative batch-service queues; characterizing the composed frontier is open and maps directly onto a Pi 5 $\rightarrow$ Orin $\rightarrow$ DGX-class testbed hierarchy.

Remark 6 (Long-context regime). With $b_{KV} > 0$ in (3), the memory floor grows with $\sum_{i \in \mathcal{B}} L_i$ and the knee $\Gamma^\dagger$ shifts right, so speculation remains optimal at larger batch sizes for long contexts; Assumption 2 captures this without modification.

Remark 7 (Position within prior art). Datacenter systems select speculation lengths to maximize a goodput or SLO objective at the *current* operating point $[?, ?, ?, ?]$, and edge systems architect distributed drafting and verification around throughput or SLO targets $[?, ?, ?, ?, ?]$. In the language of Proposition 1, all are the degenerate case $w = \delta_{x_0}$ of (9); none characterizes or optimizes the frontier, couples wireless resource allocation with speculation control, or handles censored acceptance learning with guarantees.